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Project:

## Reconstruction and Modernisation of HPP Perućica - Phase II

Title:

## Transfer functions and modelling

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# 1 Introduction

This document provides transfer functions and models which are suitable to describe Voith HyCon Digital Turbine Governors as part of computer simulation models for the electric power system.
The transfer functions are represented according to the recommended practices defined by IEEE Guide for the Application of Turbine Governing Systems for Hydroelectric Generating Units, IEEE Std 1207™-2011 and IEC Guide to specification of hydraulic turbine governing systems, IEC 61362 Ed.2.0.
It is important to note that IEEE and IEC define only simplified general models and parameters for Hydro Units governing systems.

## 2 Symbols and relationship between different notations

International standards are using different symbols for elements of governors in hydropower plants. Below is a summary of the most important symbols and relationships between notations as defined in IEC 61362 Ed.2.0, IEC 60050-351, IEEE Std. 1207-2011 and ASME PTC 29-2005.

| **Symbols and correlation** | **Description** |
|---|---|
| $K_p$ | Proportional gain (IEEE 1207, ASME PTC 29)<br>Proportional action coefficient (IEC 61362, IEC 60050-351)<br>In German: Proportionalbeiwert (IEC 61362, IEC 60050-351) |
| $b_p$ | Permanent speed droop (IEEE 1207, ASME PTC 29)<br>Permanent droop (IEC 61362) |
| $K_i = K_I = \frac{K_p}{T_i}$ | Integral gain $K_i$ (IEEE 1207)<br>Integral action coefficient $K_I$ (IEC 61362, IEC 60050-351)<br>In German: Integrierbeiwert $K_I$ (IEC 61362, IEC 60050-351)<br>Integral gain $K_I$ (ASME PTC 29) |
| $T_i = \frac{K_p}{K_I} = K_p T_I = T_d$ | Reset time $T_i$ (IEC 61362, IEC 60050-351)<br>In German: Nachstellzeit $T_i$ (IEC 61362, IEC 60050-351)<br>Damping time constant $T_d$ (ASME PTC 29) |
| $T_I = \frac{1}{K_I}$ | Integral action time (IEC 61362, IEC 60050-351)<br>In German: Integrierzeit (IEC 61362, IEC 60050-351) |
| $K_d = K_D = K_{1D} T_{1D}$ | Derivative gain $K_d$ (IEEE 1207)<br>Derivative action coefficient $K_D$ (IEC 61362, IEC 60050-351)<br>In German: Differenzierbeiwert $K_D$ (IEC 61362, IEC 60050-351) |
| $T_D = K_D$ | Derivative action time $T_D$ (IEC 61362, IEC 60050-351)<br>In German: Differenzierzeit $T_D$ (IEC 61362, IEC 60050-351) |
| $K_D = K_{1D} \neq K_{D_{IEC61362}}$ | Derivative gain $K_D$ (ASME PTC 29) |
| $T_d = \frac{K_D}{K_p} = \frac{T_D}{K_p}$ | Rate time $T_d$ (IEC 61362, IEC 60050-351)<br>In German: Vorhaltzeit $T_d$ (IEC 61362, IEC 60050-351) |
| $T_d = T_{1D}$ | Time constant of the damping device $T_d$ (IEEE 1207) |
| $T_V = T_d = T_{1D}$ | Derivative Filter Time Constant $T_V$, $T_d$ (ASME PTC 29) |

# 3 Units

The parameters and variables of the models (with the exception of head) are represented in percent with reference values according to the following table:

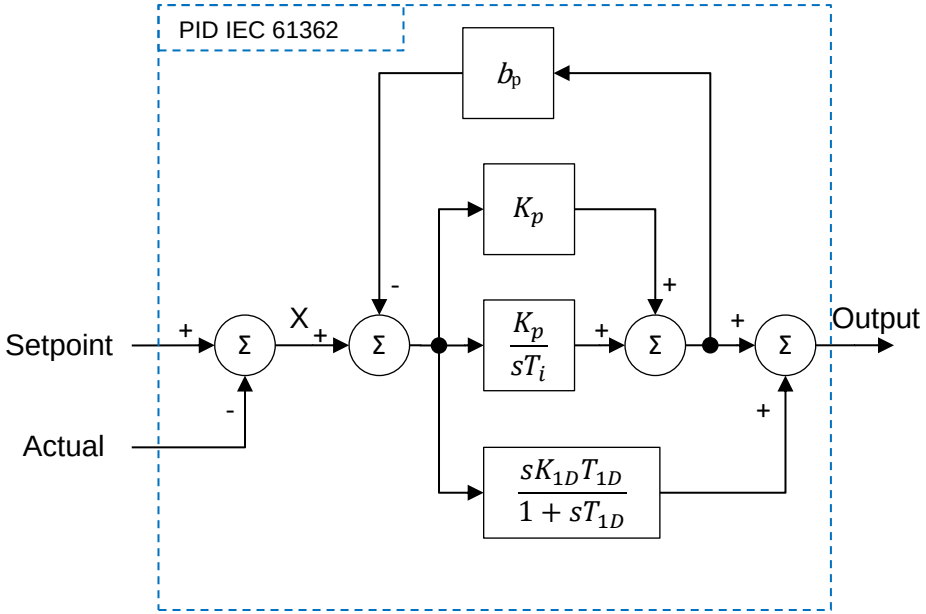
| **Values** | **Unit** | **Percent** | **Reference value** |
|---|---|---|---|
| Active Power | MW | % of rated power | Turbine rated power |
| Main Distributing Valve Position | - | % | Maximum amplitude |
| Speed | rpm | % of rated speed | Shaft rated speed |
| Grid Frequency | Hz | % of nominal frequency | Grid nominal frequency |
| Turbine Head | m | Not applicable | Turbine rated head |
| Water level | m | No general definition available | Sometimes used: max – min level |
| Units and reference values | | | |

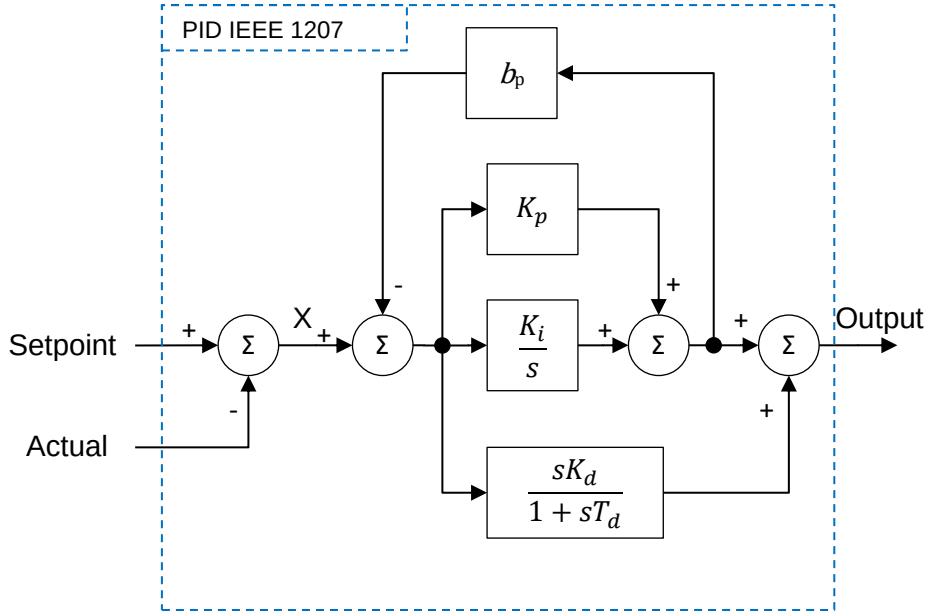
# 4 Nomenclature

| | |
|---|---|
| ACCL | Acceleration control |
| AFC | Frequency influence (automatic frequency control) |
| CAVN | Cavitation |
| DEFL | Deflector |
| $DEFL_{ONL}$ | Deflector at Speed No Load (%) |
| $DEFL_{sp}$ | Deflector Control Setpoint (%) |
| $F_{nom}$ | Nominal Grid Frequency (%) |
| $F_{act}$ | Actual Grid Frequency (%) |
| $F_{actFiltered}$ | Actual Filtered Grid Frequency (%) |
| FCR | Frequency Containment Reserve (including FCR-N and FCR-D) |
| FCR-N | Frequency Containment Reserve for Normal Operation |
| FCR-D | Frequency Containment Reserve for Disturbances |
| FCR-D up | Frequency Containment Reserve for Disturbances upwards regulation ($F_{act} < F_{nom}$) |
| FCR-D down | Frequency Containment Reserve for Disturbances downwards regulation ($F_{act} > F_{nom}$) |
| FLOW | Flow control |
| $FLOW_{act}$ | Actual Flow(%) |
| $FLOW_{ONL}$ | Flow at Speed No Load (%) |
| $FLOW_{sp}$ | Flow Control Setpoint (%) |
| $FLOW\text{-}LMT_{sp}$ | Flow Limitation Setpoint (%) |
| FSML | Frequency Sensitive Mode and Limited Frequency Sensitive Mode |
| FSM | Frequency Sensitive Mode |
| LFSM | Limited Frequency Sensitive Mode |
| LFSM-U | Limited Frequency Sensitive Mode – Underfrequency ($F_{act} < F_{nom}$) |
| LFSM-O | Limited Frequency Sensitive Mode – Overfrequency ($F_{act} > F_{nom}$) |
| MV | Actual Main Distributing Valve position (%) |
| NOZ | Nozzle |
| OVRLD | Overload |
| POSN | Position control |
| POW | Power control |
| $POW_{sp}$ | Power Control Setpoint (%) |

| | |
|---|---|
| $POW_{act}$ | Actual Power (%) |
| POW-FF | Power control feed-forward curve |
| $POW\text{-}FLOW_{sp}$ | Flow Setpoint from Power Control (Power Control Loop Output) (%) |
| POW-LMT | Power limit control |
| $POW\text{-}LMT_{sp}$ | Power Limitation Setpoint (%) |
| PP | Pump operation |
| RB | Runner blade |
| SC | Synchronous condenser |
| SPD | Speed control |
| $SPD_{sp}$ | Speed Control Setpoint (%) |
| $SPD_{act}$ | Actual Speed (%) |
| SPD-NO-LD | Speed control no-load |
| SPD-ISOL | Speed control isolated grid |
| SPD-GD | Speed control large grid |
| SPD-BS | Speed control blackstart |
| $SPD\text{-}DEFL_{sp}$ | Deflector Setpoint from Speed Control (Speed Control Loop Output) (%) |
| $SPD\text{-}FLOW_{sp}$ | Flow Setpoint from Speed Control (Speed Control Loop Output) (%) |
| SRG | Surge control |
| WG | Wicket gate |
| HyCon Governor Nomenclature | |

# 5 General PID structure




Note: PID structure is different concerning permanent speed droop and derivative part compared to example structure of figure 9 of IEEE 1207-2011

| Parameters | | Comments |
|---|---|---|
| $bp$ | Permanent droop | IEC 61362, IEEE1207 |
| $K_p$ | Proportional gain | IEC 61362, IEEE1207 |
| $T_i$ | Reset time | IEC 61362 |
| $K_{1D}$ | Derivative gain | IEC 61362 |
| $T_{1D}$ | Derivative time constant | IEC 61362 |
| $K_i$ | Integral gain | IEEE1207 |
| $K_d$ | Derivative gain | IEEE1207 |
| $T_d$ | Time constant of the damping device | IEEE1207 |
| General PID parameters | | |

Conversion rules:

$b_{p\,\mathrm{IEEE1207}} = b_{p\,\mathrm{IEC61362}}$

$K_{p\,\mathrm{IEEE1207}} = K_{p\,\mathrm{IEC61362}}$

$K_{i\,\mathrm{IEEE1207}} = K_{p\,\mathrm{IEC61362}} / T_{i\,\mathrm{IEC61362}}$

$K_{d\,\mathrm{IEEE1207}} = K_{1D\,\mathrm{IEC61362}} * T_{1D\,\mathrm{IEC61362}}$

$T_{d\,\mathrm{IEEE1207}} = T_{1D\,\mathrm{IEC61362}}$

Transfer function PID according to IEC 61362 nomenclature

$$\frac{Output}{X} = \frac{K_p + \frac{K_p}{s * T_i} + \frac{s * K_{1D} * T_{1D}}{1 + s * T_{1D}}}{1 + b_p \left(K_p + \frac{K_p}{s * T_i}\right)}$$

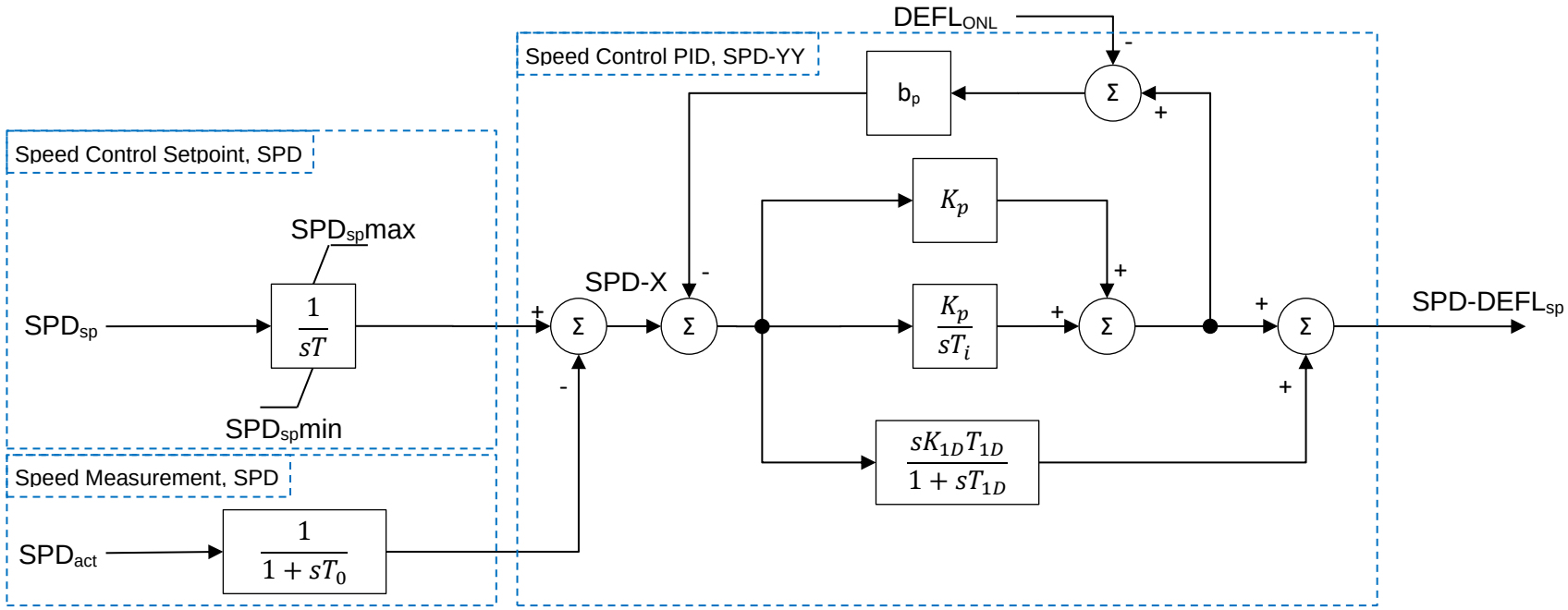
Transfer function PID according to IEEE 1207 nomenclature

$$\frac{Output}{X} = \frac{K_p + \frac{K_i}{s} + \frac{s * K_d}{1 + s * T_d}}{1 + b_p \left(K_p + \frac{K_i}{s}\right)}$$

Note: PID structure is different concerning permanent speed droop and derivative part compared to example structure of figure 9 of IEEE 1207-2011.

# 6 Transfer functions according to IEC 61362 nomenclature

## 6.1 Speed Control Mode

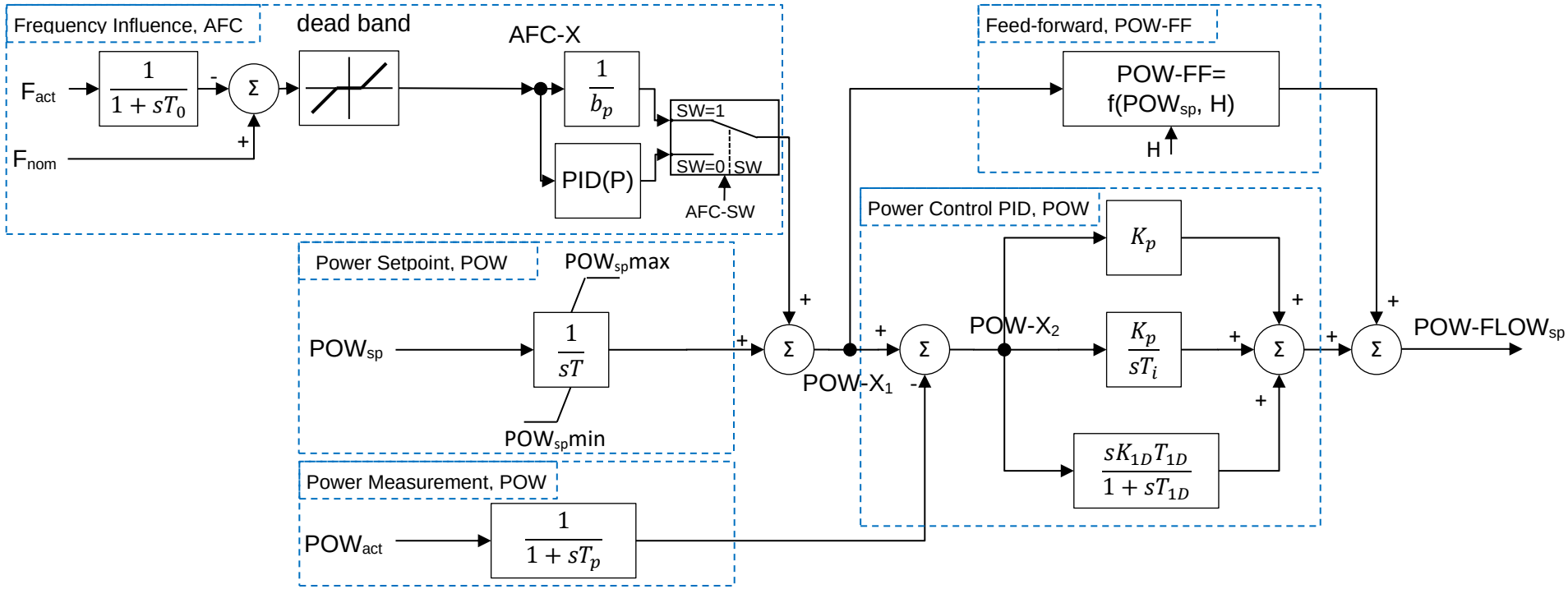


| Parameters | | Parameter range | Typical | Comments |
|---|---|---|---|---|
| $SPD_{sp}min$ | Minimum allowed Speed set point | 0% - 200% | 90% | |
| $SPD_{sp}max$ | Maximum allowed Speed set point | 0% - 200% | 110% | |
| *SPD-T* | Setpoint ramp time 0%-100% for Speed Control mode [s] | 10 -120s | 60s | |
| $SPD\text{-}T_0$ | Speed Measurement Filter Time Constant [s] | 0.02 – 1s | 0.02s | |
| $SPD\text{-}YY\text{-}b_p$ | Permanent droop | 0 – 0.1 (0% - 10%) | 0 (0%) – No Load (YY= NO-LD)<br>0.03 (3%) – Isolated (YY= ISOL)<br>0.05 (5%) – Blackstart (*YY*= BS) | |
| $SPD\text{-}YY\text{-}K_p$ | Proportional gain | 0.1 – 20 | 1.2 – No Load (*YY*= NO-LD)<br>0.9 – Isolated (*YY*= ISOL)<br>0.8 – Blackstart (*YY*= BS) | |
| $SPD\text{-}YY\text{-}T_i$ | Reset time [s] | 0.1 – 100s | 6.0s – No Load (*YY*= NO-LD)<br>7.0s – Isolated (*YY*= ISOL)<br>5.0s – Blackstart (*YY*= BS) | |
| $SPD\text{-}YY\text{-}K_{1D}$ | Derivative gain | 0 – 10 | 0.0 – No Load (*YY*= NO-LD)<br>0.0 – Isolated (*YY*= ISOL)<br>0.0 – Blackstart (*YY*= BS) | |
| $SPD\text{-}YY\text{-}T_{1D}$ | Derivative time constant [s] | 0.1 – 10s | 2s (*YY*= NO-LD, ISOL, BS) | |
| Speed control mode parameters | | | | |

Transfer function Speed Control PID ($DEFL_{ONL}$ = constant):

$$\frac{SPD\text{-}DEFL_{sp}}{SPD\text{-}X} = \frac{SPD\text{-}K_p + \frac{SPD\text{-}K_p}{s * SPD\text{-}T_i} + \frac{s * SPD\text{-}K_{1D} * SPD\text{-}T_{1D}}{1 + s * SPD\text{-}T_{1D}}}{1 + SPD\text{-}b_p \left( SPD\text{-}K_p + \frac{SPD\text{-}K_p}{s * SPD\text{-}T_i} \right)}$$

## 6.2 Power control mode



| Parameters | | Parameter Range | Typical | Comments |
|---|---|---|---|---|
| $POW_{sp}min$ | Minimum allowed Power set point | 0 – 120% | 0% | |
| $POW_{sp}max$ | Maximum allowed Power set point | 0 – 120% | 100% | |
| $AFC\text{-}T_0$ | Frequency Measurement Filter Time Constant [s] | 0.02 – 1s | 0.02s | |
| $AFC\text{-}b_p$ | Frequency influence (droop) bp | 0.001 – 0.1<br>(0.1% - 10%) | 0.05 (5%) | |
| $POW\text{-}T_p$ | Power Measurement Filter Time Constant [s] | 0.02 – 1s | 0.2s | |
| $POW\text{-}T$ | Setpoint ramp time 0%-100% for Power Control mode [s] | 10 – 120s | 60s | |
| $POW\text{-}K_p$ | Proportional gain | 0.1 – 20 | 0.1 | |
| $POW\text{-}T_i$ | Reset time [s] | 0.1 – 100s | 6s | |
| $POW\text{-}K_{1D}$ | Derivative gain | 0 – 10 | 0 | |
| $POW\text{-}T_{1D}$ | Derivative time constant [s] | 0.1 – 10s | 1s | |
| POW-FF | Corresponding opening to Head and Power setpoint (=dy/dP for actual operating point) | | | |
| AFC-SW | Frequency influence software switch | 0 - 1 | 1 | 0 = *PID*<br>1 = bp |
| Power control mode parameters | | | | |

Transfer function for evaluation of the response to power disturbances

$$\frac{POW\text{-}FLOW_{sp}}{POW\text{-}X_2} = POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}$$

$$\frac{POW\text{-}FLOW_{sp}}{POW_{act}} = \frac{-1}{1 + s * POW\text{-}T_p} * \left(POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

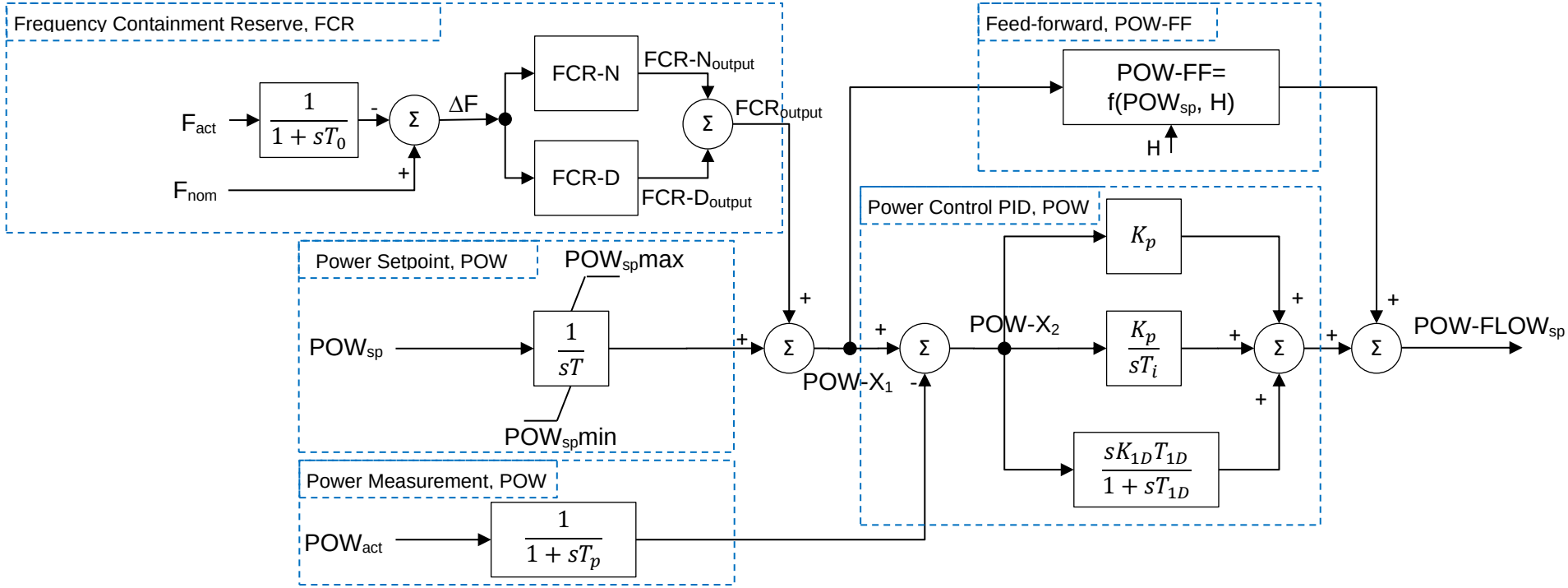
Transfer function for evaluation of the response to frequency disturbances for frequency influence with static droop

$$\frac{POW\text{-}FLOW_{sp}}{AFC\text{-}X} = \frac{1}{AFC\text{-}b_p} * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

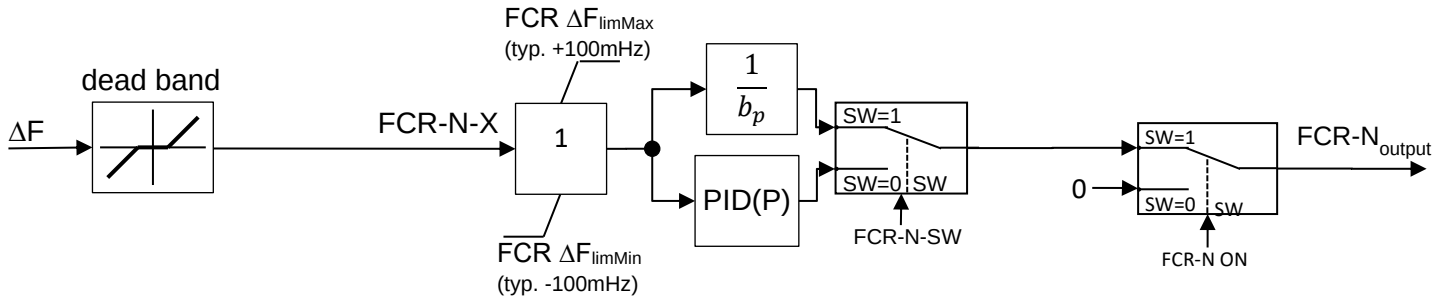
Transfer function for evaluation of the response to frequency disturbances for frequency influence with PID

$$\frac{POW\text{-}FLOW_{sp}}{AFC\text{-}X} = \frac{AFC\text{-}K_p + \frac{AFC\text{-}K_p}{s * AFC\text{-}T_i} + \frac{s * AFC\text{-}K_{1D} * AFC\text{-}T_{1D}}{1 + s * AFC\text{-}T_{1D}}}{1 + AFC\text{-}b_p * \left(AFC\text{-}K_p + \frac{AFC\text{-}K_p}{s * AFC\text{-}T_i}\right)} * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

## 6.3 Power control mode with FCR

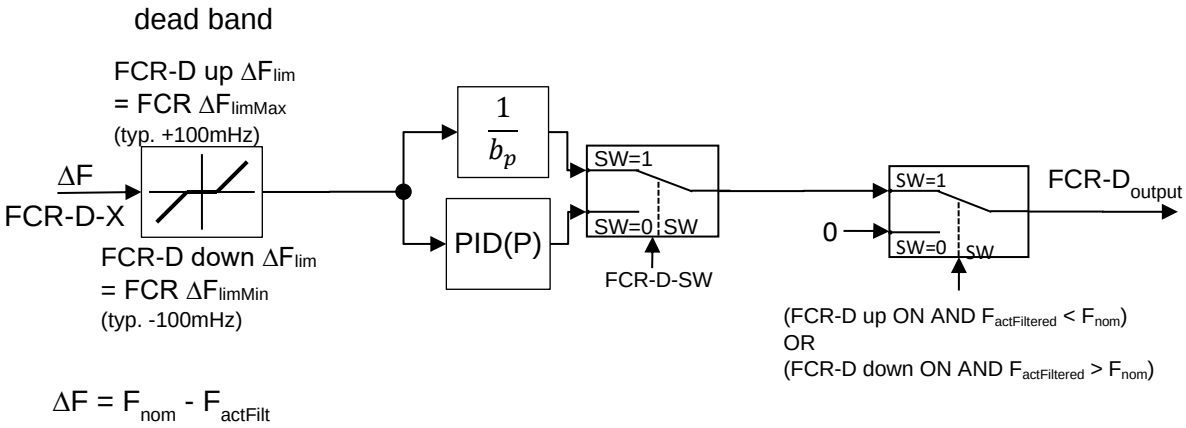


Frequency Containment Reserve Normal Operation, FCR-N




$\Delta F = F_{nom} - F_{actFilt}$

Frequency Containment Reserve for Disturbances, FCR-D



| Parameters | | Parameter Range | Typical | Comments |
|---|---|---|---|---|
| $POW_{sp}min$ | Minimum allowed Power set point | 0 – 120% | 0% | |
| $POW_{sp}max$ | Maximum allowed Power set point | 0 – 120% | 100% | |
| $FCR\text{-}T_0$ | Frequency Measurement Filter Time Constant [s] | 0.02 – 1s | 0.02s | |
| $FCR\text{-}N\text{-}b_p$ | FCR-N (droop) bp | 0.001 – 0.1<br>(0.1% - 10%) | 0.05 (5%) | |
| $FCR\text{-}D\text{-}b_p$ | FCR-D (droop) bp | 0.001 – 0.1<br>(0.1% - 10%) | 0.05 (5%) | |
| $POW\text{-}T_p$ | Power Measurement Filter Time Constant [s] | 0.02 – 1s | 0.2s | |
| $POW\text{-}T$ | Setpoint ramp time 0%-100% for Power Control mode [s] | 10 – 120s | 60s | |
| $POW\text{-}K_p$ | Proportional gain | 0.1 – 20 | 0.1 | |
| $POW\text{-}T_i$ | Reset time [s] | 0.1 – 100s | 6s | |
| $POW\text{-}K_{1D}$ | Derivative gain | 0 – 10 | 0 | |
| $POW\text{-}T_{1D}$ | Derivative time constant [s] | 0.1 – 10s | 1s | |
| POW-FF | Corresponding opening to Head and Power setpoint (=dy/dP for actual operating point) | | | |
| FCR-N-SW | FCR-N software switch PID/droop | 0 - 1 | 0 | 0 = *PID*<br>1 = bp |
| FCR-D-SW | FCR-D software switch PID/droop | 0 - 1 | 0 | 0 = *PID*<br>1 = bp |
| Power control mode with FCR-N and FCR-D parameters | | | | |

Transfer function for evaluation of the response to power disturbances

$$\frac{POW\text{-}FLOW_{sp}}{POW\text{-}X_2} = POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}$$

$$\frac{POW\text{-}FLOW_{sp}}{POW_{act}} = \frac{-1}{1 + s * POW\text{-}T_p} * \left(POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

Transfer function for evaluation of the response to frequency disturbances for FCR-N with static droop ($|\Delta F| \leq |FCR\ \Delta F_{lim}|$)

$$\frac{POW\text{-}FLOW_{sp}}{FCR\text{-}N\text{-}X} = \frac{1}{FCR\text{-}N\text{-}b_p} * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

Transfer function for evaluation of the response to frequency disturbances for FCR-N with PID ($|\Delta F| \leq |FCR\ \Delta F_{lim}|$)

$$\frac{POW\text{-}FLOW_{sp}}{FCR\text{-}N\text{-}X} = \frac{FCR\text{-}N\text{-}K_p + \frac{FCR\text{-}N\text{-}K_p}{s * FCR\text{-}N\text{-}T_i} + \frac{s * FCR\text{-}N\text{-}K_{1D} * FCR\text{-}N\text{-}T_{1D}}{1 + s * FCR\text{-}N\text{-}T_{1D}}}{1 + FCR\text{-}N\text{-}b_p * \left(FCR\text{-}N\text{-}K_p + \frac{FCR\text{-}N\text{-}K_p}{s * FCR\text{-}N\text{-}T_i}\right)} * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

Transfer function for evaluation of the response to frequency disturbances for FCR-D with static droop ($|\Delta F| > |FCR\ \Delta F_{lim}|$)

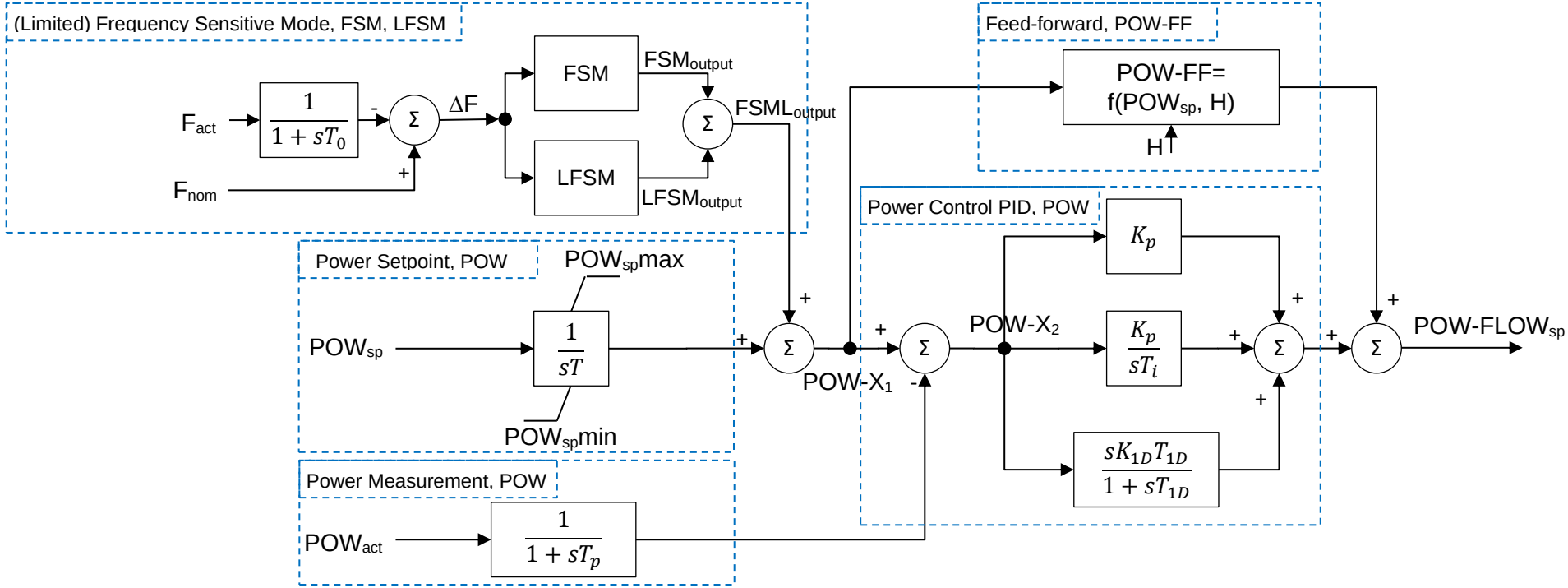
$$\frac{POW\text{-}FLOW_{sp}}{FCR\text{-}D\text{-}X} = \left[\frac{1}{FCR\text{-}N\text{-}b_p} * \frac{FCR\Delta F_{lim}}{FCR\text{-}D\text{-}X} + \frac{1}{FCR\text{-}D\text{-}b_p} * \left(1 - \frac{FCR\Delta F_{lim}}{FCR\text{-}D\text{-}X}\right)\right] * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

Transfer function for evaluation of the response to frequency disturbances for FCR-D with PID ($|\Delta F| > |FCR\ \Delta F_{lim}|$)

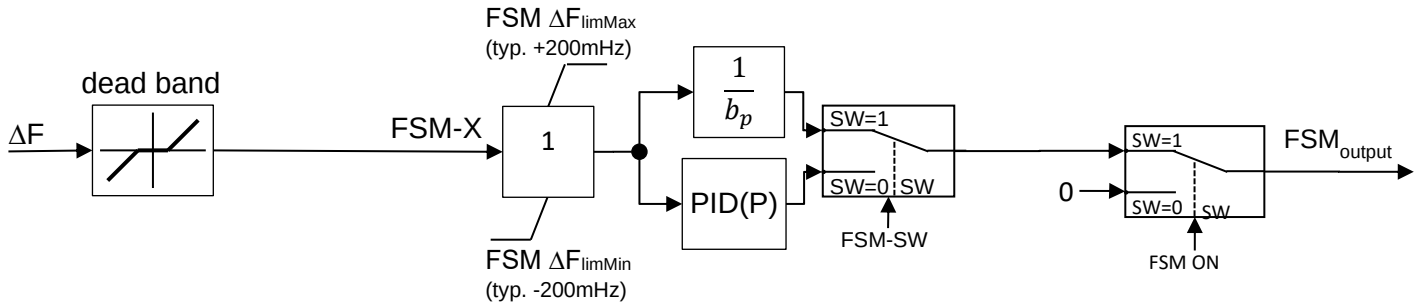
$$\frac{POW\text{-}FLOW_{sp}}{FCR\text{-}D\text{-}X} = \left[\frac{1}{FCR\text{-}N\text{-}b_p} * \frac{FCR\Delta F_{lim}}{FCR\text{-}D\text{-}X} + \frac{FCR\text{-}D\text{-}K_p + \frac{FCR\text{-}D\text{-}K_p}{s * FCR\text{-D-}T_i} + \frac{s * FCR\text{-}D\text{-}K_{1D} * FCR\text{-}D\text{-}T_{1D}}{1 + s * FCR\text{-}D\text{-}T_{1D}}}{1 + FCR\text{-}D\text{-}b_p * \left(FCR\text{-}D\text{-}K_p + \frac{FCR\text{-}D\text{-}K_p}{s * FCR\text{-}D\text{-}T_i}\right)} * \left(1 - \frac{FCR\Delta F_{lim}}{FCR\text{-}D\text{-}X}\right)\right]$$

$$* \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

## 6.4 Power control mode FSM, LFSM

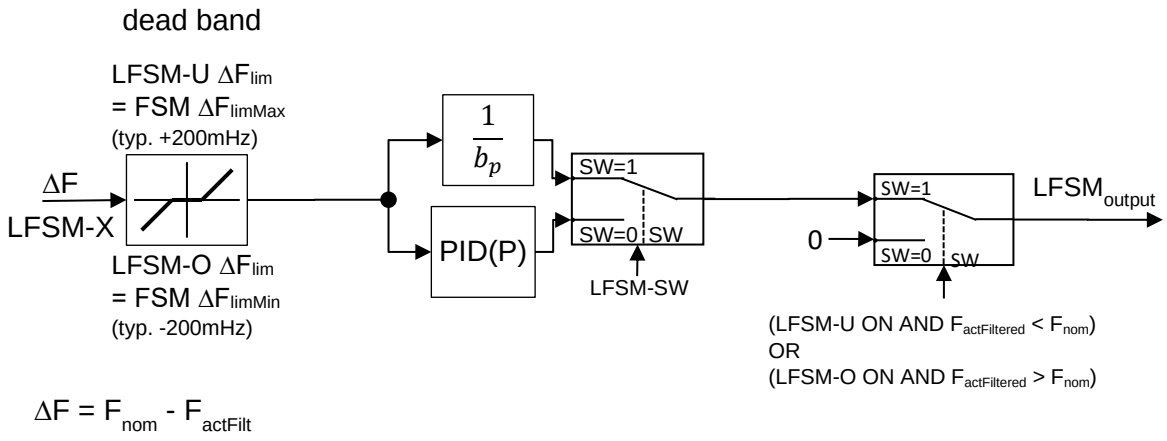


Frequency Sensitive Mode, FSM




$\Delta F = F_{nom} - F_{actFilt}$

Limited Frequency Sensitive Mode, LFSM



| Parameters | | Parameter Range | Typical | Comments |
|---|---|---|---|---|
| $POW_{sp}min$ | Minimum allowed Power set point | 0 – 120% | 0% | |
| $POW_{sp}max$ | Maximum allowed Power set point | 0 – 120% | 100% | |
| $FSM\text{-}T_0$ | Frequency Measurement Filter Time Constant [s] | 0.02 – 1s | 0.02s | |
| $FSM\text{-}b_p$ | FSM (droop) bp | 0.001 – 0.1 (0.1% - 10%) | 0.05 (5%) | |
| $LFSM\text{-}b_p$ | LFSM (droop) bp | 0.001 – 0.1 (0.1% - 10%) | 0.05 (5%) | |
| $POW\text{-}T_p$ | Power Measurement Filter Time Constant [s] | 0.02 – 1s | 0.2s | |
| $POW\text{-}T$ | Setpoint ramp time 0%-100% for Power Control mode [s] | 10 – 120s | 60s | |
| $POW\text{-}K_p$ | Proportional gain | 0.1 – 20 | 0.1 | |
| $POW\text{-}T_i$ | Reset time [s] | 0.1 – 100s | 6s | |
| $POW\text{-}K_{1D}$ | Derivative gain | 0 – 10 | 0 | |
| $POW\text{-}T_{1D}$ | Derivative time constant [s] | 0.1 – 10s | 1s | |
| POW-FF | Corresponding opening to Head and Power setpoint (=dy/dP for actual operating point) | | | |
| FSM-SW | FSM software switch PID/droop | 0 - 1 | 0 | 0 = *PID*<br>1 = bp |
| LFSM-SW | LFSM software switch PID/droop | 0 - 1 | 0 | 0 = *PID*<br>1 = bp |
| Power control mode with FSM/LFSM parameters | | | | |

Transfer function for evaluation of the response to power disturbances

$$\frac{POW\text{-}FLOW_{sp}}{POW\text{-}X_2} = POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}$$

$$\frac{POW\text{-}FLOW_{sp}}{POW_{act}} = \frac{-1}{1 + s * POW\text{-}T_p} * \left(POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

Transfer function for evaluation of the response to frequency disturbances for FSM with static droop ($|\Delta F| \leq |FSM\ \Delta F_{lim}|$)

$$\frac{POW\text{-}FLOW_{sp}}{FSM\text{-}X} = \frac{1}{FSM\text{-}b_p} * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

Transfer function for evaluation of the response to frequency disturbances for FSM with PID ($|\Delta F| \leq |FSM\ \Delta F_{lim}|$)

$$\frac{POW\text{-}FLOW_{sp}}{FSM\text{-}X} = \frac{FSM\text{-}K_p + \frac{FSM\text{-}K_p}{s * FSM\text{-}T_i} + \frac{s * FSM\text{-}K_{1D} * FSM\text{-}T_{1D}}{1 + s * FSM\text{-}T_{1D}}}{1 + FSM\text{-}b_p * \left(FSM\text{-}K_p + \frac{FSM\text{-}K_p}{s * FSM\text{-}T_i}\right)} * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

Transfer function for evaluation of the response to frequency disturbances for LFSM with static droop ($|\Delta F| > |FSM\ \Delta F_{lim}|$)

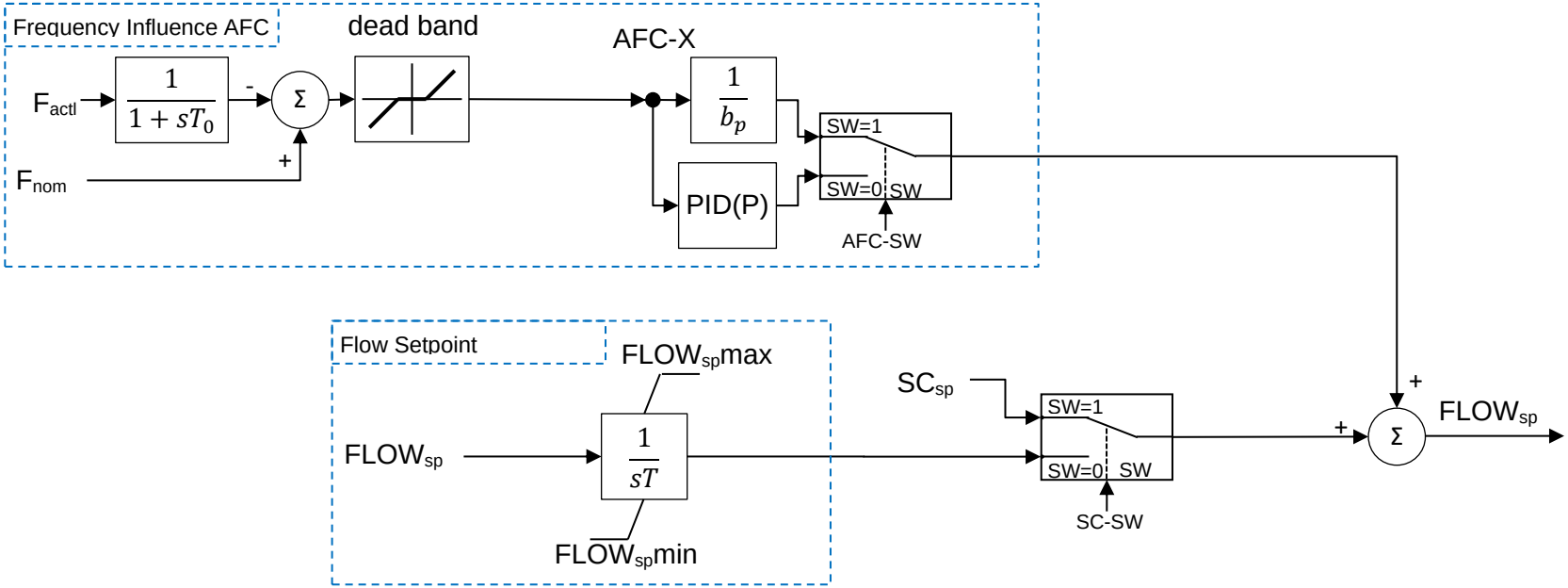
$$\frac{POW\text{-}FLOW_{sp}}{LFSM\text{-}X} = \left[\frac{1}{FSM\text{-}b_p} * \frac{FSM\Delta F_{lim}}{LFSM\text{-}X} + \frac{1}{LFSM\text{-}b_p} * \left(1 - \frac{FSM\Delta F_{lim}}{LFSM\text{-}X}\right)\right] * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

Transfer function for evaluation of the response to frequency disturbances for LFSM with PID ($|\Delta F| > |FSM\ \Delta F_{lim}|$)

$$\frac{POW\text{-}FLOW_{sp}}{LFSM\text{-}X} = \left[\frac{1}{FSM\text{-}b_p} * \frac{FSM\Delta F_{lim}}{LFSM\text{-}X} + \frac{LFSM\text{-}K_p + \frac{LFSM\text{-}K_p}{s * LFSM\text{-}T_i} + \frac{s * LFSM\text{-}K_{1D} * LFSM\text{-}T_{1D}}{1 + s * LFSM\text{-}T_{1D}}}{1 + LFSM\text{-}b_p * \left(LFSM\text{-}K_p + \frac{LFSM\text{-}K_p}{s * LFSM\text{-}T_i}\right)} * \left(1 - \frac{FSM\Delta F_{lim}}{LFSM\text{-}X}\right)\right]$$

$$* \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

## 6.5 Flow control mode



| Parameters | | Parameter Range | Typical | Comments |
|---|---|---|---|---|
| $FLOW_pmin$ | Minimum allowed Flow set point | 0 – 100% | 0% | |
| $FLOW_{sp}max$ | Maximum allowed Flow set point | 0 – 100% | 100% | |
| *FLOW-T* | Setpoint ramp time 0%-100% for Flow Control mode [s] | 10 – 120s | 60s | |
| *AFC-*$T_0$ | Speed Measurement Filter Time Constant [s] | 0.02 – 1s | 0.02s | |
| *AFC-*$b_p$ | Frequency influence (droop) bp | 0.001 – 0.1<br>(0.1% – 10%) | 0.05 (5%) | |
| AFC-SW | Frequency influence software switch | 0 – 1 | 1 | 0 = PID<br>1 = bp |
| SC-SW | Synchronous condenser software switch | 0 – 1 | 0 | 0 = Off<br>1 = On |
| $SC_{sp}$ | Flow set point in Synchronous condenser mode | 0 – 100% | 0% | |
| Flow control mode parameters | | | | |

**Note**: AFC can also stand for FCR (see chapter *Power control mode with FCR*) or FSML (see chapter *Power control mode with FSM*)

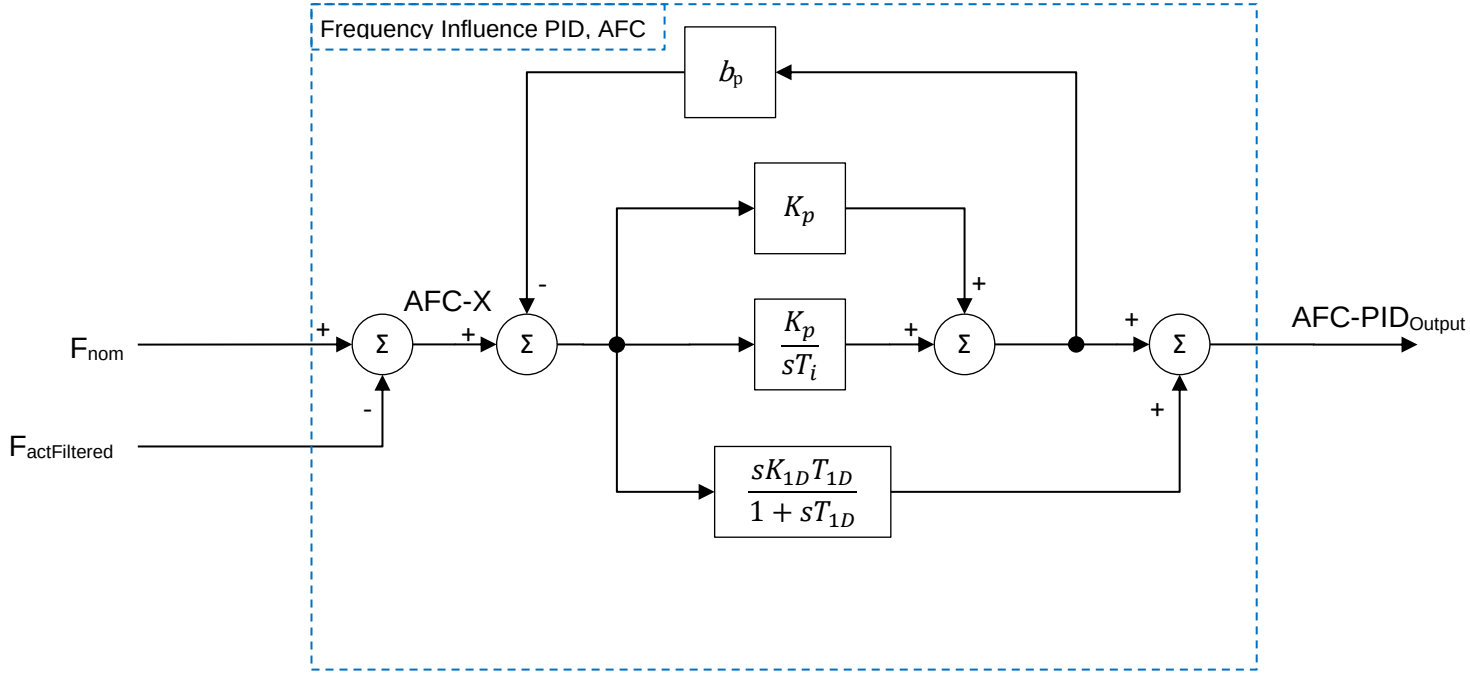
Transfer function for evaluation of the response to frequency disturbances for frequency influence with static droop

$$\frac{\mathrm{FLOW_{sp}}}{AFC\text{-}X} = \frac{1}{AFC\text{-}b_p}$$

Transfer function for evaluation of the response to frequency disturbances for frequency influence with PID

$$\frac{\mathrm{FLOW_{sp}}}{AFC\text{-}X} = \frac{AFC\text{-}K_p + \frac{AFC\text{-}K_p}{s * AFC\text{-}T_i} + \frac{s * AFC\text{-}K_{1D} * AFC\text{-}T_{1D}}{1 + s * AFC\text{-}T_{1D}}}{1 + AFC\text{-}b_p \left( AFC\text{-}K_p + \frac{AFC\text{-}K_p}{s * AFC\text{-}T_i} \right)}$$

## 6.6 Frequency influence PID loop (including FCR mode)



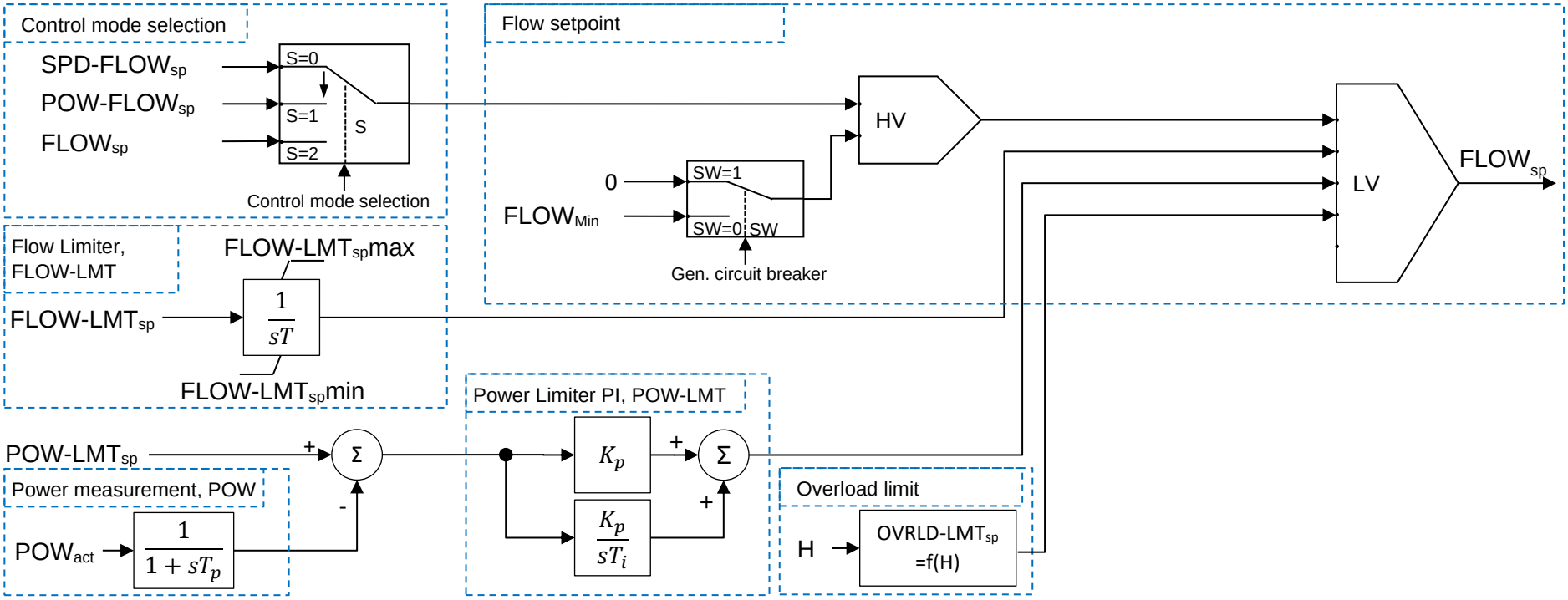
| Parameters | | Parameter Range | Typical | Comments |
|---|---|---|---|---|
| $AFC\text{-}bp$ | Permanent droop | 0 – 0.1 (0% - 10%) | 0.05 (5%) | |
| $AFC\text{-}K_p$ | Proportional gain | 0.1 – 20 | 1.5 | |
| $AFC\text{-}T_i$ | Reset time [s] | 0.1 – 100s | 3s | |
| $AFC\text{-}K_{1D}$ | Derivative gain | 0 – 10 | 0 | |
| $AFC\text{-}T_{1D}$ | Derivative time constant [s] | 0.1 – 10s | 0s | |
| Frequency influence PID parameters | | | | |

Transfer function Frequency Influence PID:

$$\frac{AFC\text{-}PID_{Output}}{AFC\text{-}X} = \frac{AFC\text{-}K_p + \frac{AFC\text{-}K_p}{s * AFC\text{-}T_i} + \frac{s * AFC\text{-}K_{1D} * AFC\text{-}T_{1D}}{1 + s * AFC\text{-}T_{1D}}}{1 + AFC\text{-}b_p\left(AFC\text{-}K_p + \frac{AFC\text{-}K_p}{s * AFC\text{-}T_i}\right)}$$

**Note**: AFC can also stand for FCR-N and FCR-D (see chapter *Power control mode with FCR*) or FSM and LFSM (see chapter *Power control mode with FSM, LFSM*)

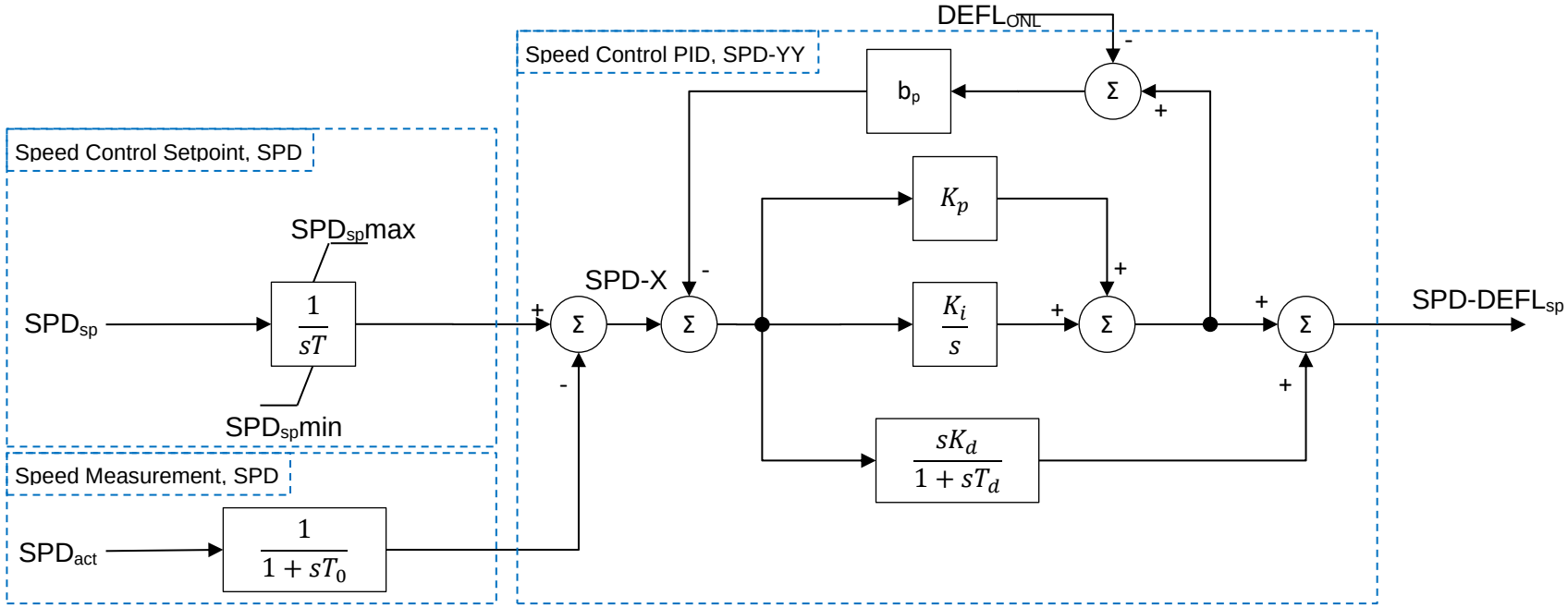
## 6.7 Control signal selection



| Parameters | | Parameter Range | Typical | Comments |
|---|---|---|---|---|
| FLOW-LMT$_{sp}$min | Minimum allowed Flow Limiter set point | 0 - 100% | 0% | |
| FLOW-LMT$_{sp}$max | Maximum allowed Flow Limiter set point | 0 - 100% | 100% | |
| *FLOW-LMT-T* | Setpoint ramp time 0%-100% for Flow Limiter [s] | 10 – 120s | 60s | |
| *POW-$T_p$* | Power Measurement Filter Time Constant [s] | 0.02 – 1s | 0.2s | |
| FLOW$_{Min}$ | Minimum flow with generator circuit breaker closed | 0 – 10% | 2% + $O_{NL}$ | |
| *POW-LMT-$K_p$* | Proportional gain – Power Limiter | 0.1 – 20 | 0.8 | |
| *POW-LMT-$T_i$* | Reset time – Power Limiter [s] | 0.1 – 100s | 6s | |
| POW-LMT$_{sp}$ | Maximum Power Limit | 0 - 100% | 100% | |
| OVRLD-LMT$_{sp}$ | Corresponding overload flow limit to Head H | - | | 11 interpolated points |
| H | Actual head | - | | |
| LV | Low value (Lowest value is transferred to output) | - | | |
| HV | High value (Highest value is transferred to output) | - | | |
| Control signal selection parameters | | | | |

# 7 Transfer functions according to IEEE 1207 nomenclature

## 7.1 Speed Control Mode

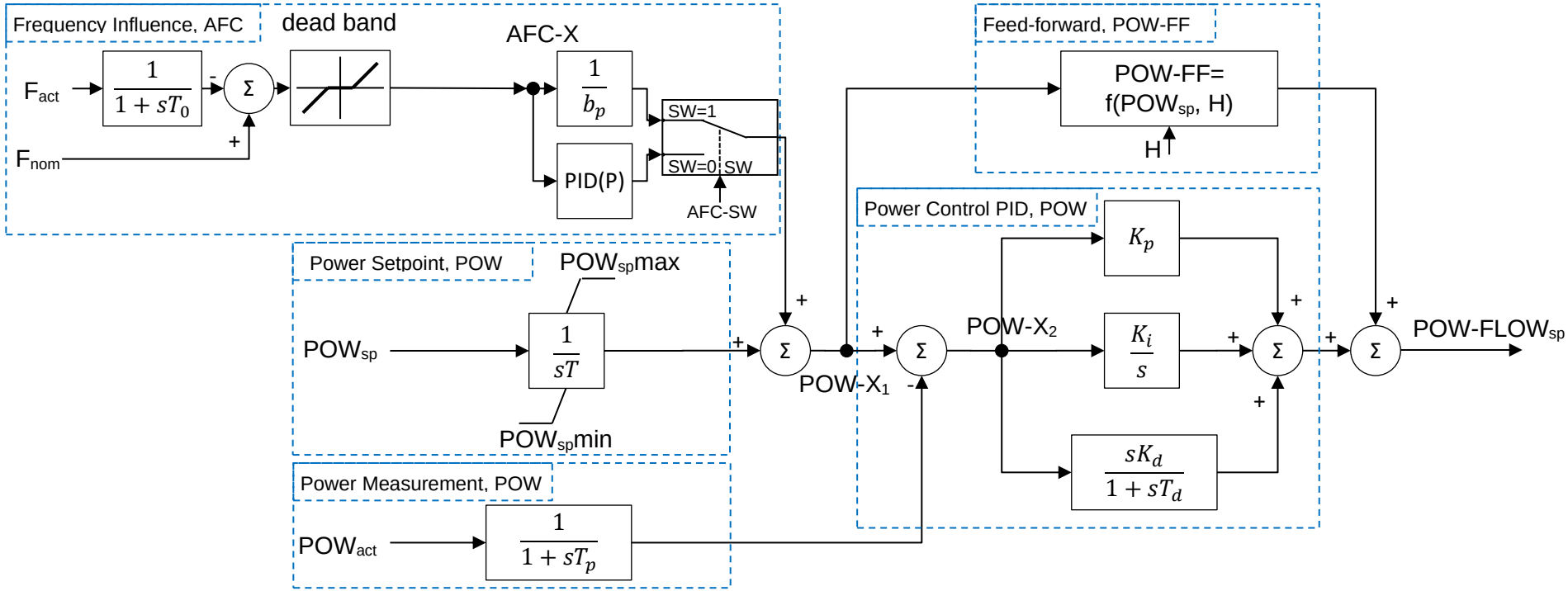


| Parameters | | Parameter range | Typical | Comments |
|---|---|---|---|---|
| $SPD_{sp}min$ | Minimum allowed Speed set point | 0% - 200% | 90% | |
| $SPD_{sp}max$ | Maximum allowed Speed set point | 0% - 200% | 110% | |
| *SPD-T* | Setpoint ramp time 0%-100% for Speed Control mode [s] | 10 -120s | 60s | |
| $SPD\text{-}T_0$ | Speed Measurement Filter Time Constant [s] | 0.02 – 1s | 0.02s | |
| $SPD\text{-}YY\text{-}b_p$ | Permanent droop | 0 – 0.1 (0% - 10%) | 0 (0%) – No Load (YY= NO-LD)<br>0.03 (3%) – Isolated (YY= ISOL)<br>0.05 (5%) – Blackstart (*YY*= BS) | |
| $SPD\text{-}YY\text{-}K_p$ | Proportional gain | 0.1 – 20 | 1.2 – No Load (*YY*= NO-LD)<br>0.9 – Isolated (*YY*= ISOL)<br>0.8 – Blackstart (*YY*= BS) | |
| $SPD\text{-}YY\text{-}K_i$ | Integral gain | 0.01 – 100 | 0.2 – No Load (*YY*= NO-LD)<br>0.13 – Isolated (*YY*= ISOL)<br>0.16 – Blackstart (*YY*= BS) | |
| $SPD\text{-}YY\text{-}K_d$ | Derivative gain | 0 – 10 | 0.0 – No Load (*YY*= NO-LD)<br>0.0 – Isolated (*YY*= ISOL)<br>0.0 – Blackstart (*YY*= BS) | |
| $SPD\text{-}YY\text{-}T_d$ | Time constant of the damping device [s] | 0.1 – 10s | 2s (*YY*= NO-LD, ISOL, BS) | |
| Speed control mode parameters | | | | |

Transfer function Speed Control PID ($DEFL_{ONL}$ = constant):

$$\frac{SPD\text{-}DEFL_{sp}}{SPD\text{-}X} = \frac{SPD\text{-}K_p + \frac{SPD\text{-}K_i}{s} + \frac{s * SPD\text{-}K_d}{1 + s * SPD\text{-}T_d}}{1 + SPD\text{-}b_p \left(SPD\text{-}K_p + \frac{SPD\text{-}K_i}{s}\right)}$$

## 7.2 Power Control Mode



| Parameters | | Parameter Range | Typical | Comments |
|---|---|---|---|---|
| $POW_{sp}min$ | Minimum allowed Power set point | 0 – 120% | 0% | |
| $POW_{sp}max$ | Maximum allowed Power set point | 0 – 120% | 100% | |
| $AFC\text{-}T_0$ | Frequency Measurement Filter Time Constant [s] | 0.02 – 1s | 0.02s | |
| $AFC\text{-}b_p$ | Frequency influence (droop) bp | 0.001 – 0.1<br>(0.1% - 10%) | 0.05 (5%) | |
| $POW\text{-}T_p$ | Power Measurement Filter Time Constant [s] | 0.02 – 1s | 0.2s | |
| $POW\text{-}T$ | Setpoint ramp time 0%-100% for Power Control mode [s] | 10 – 120s | 60s | |
| $POW\text{-}K_p$ | Proportional gain | 0.1 – 20 | 0.1 | |
| $POW\text{-}K_i$ | Integral gain | 0.01 – 100 | 0.02 | |
| $POW\text{-}K_d$ | Derivative gain | 0 – 10 | 0 | |
| $POW\text{-}T_d$ | Time constant of the damping device [s] | 0.1 – 10s | 1s | |
| POW-FF | Corresponding opening to Head and Power setpoint (=dy/dP for actual operating point) | | | |
| AFC-SW | Frequency influence software switch | 0 - 1 | 1 | 0 = *PID*<br>1 = bp |
| Power control mode parameters | | | | |

Transfer function for evaluation of the response to power disturbances

$$\frac{POW\text{-}FLOW_{sp}}{POW\text{-}X_2} = POW\text{-}K_p + \frac{POW\text{-}K_i}{s} + \frac{s * POW\text{-}K_d}{1 + s * POW\text{-}T_d}$$

$$\frac{POW\text{-}FLOW_{sp}}{POW_{act}} = \frac{-1}{1 + s * POW\text{-}T_p} * \left(POW\text{-}K_p + \frac{POW\text{-}K_i}{s} + \frac{s * POW\text{-}K_d}{1 + s * POW\text{-}T_d}\right)$$

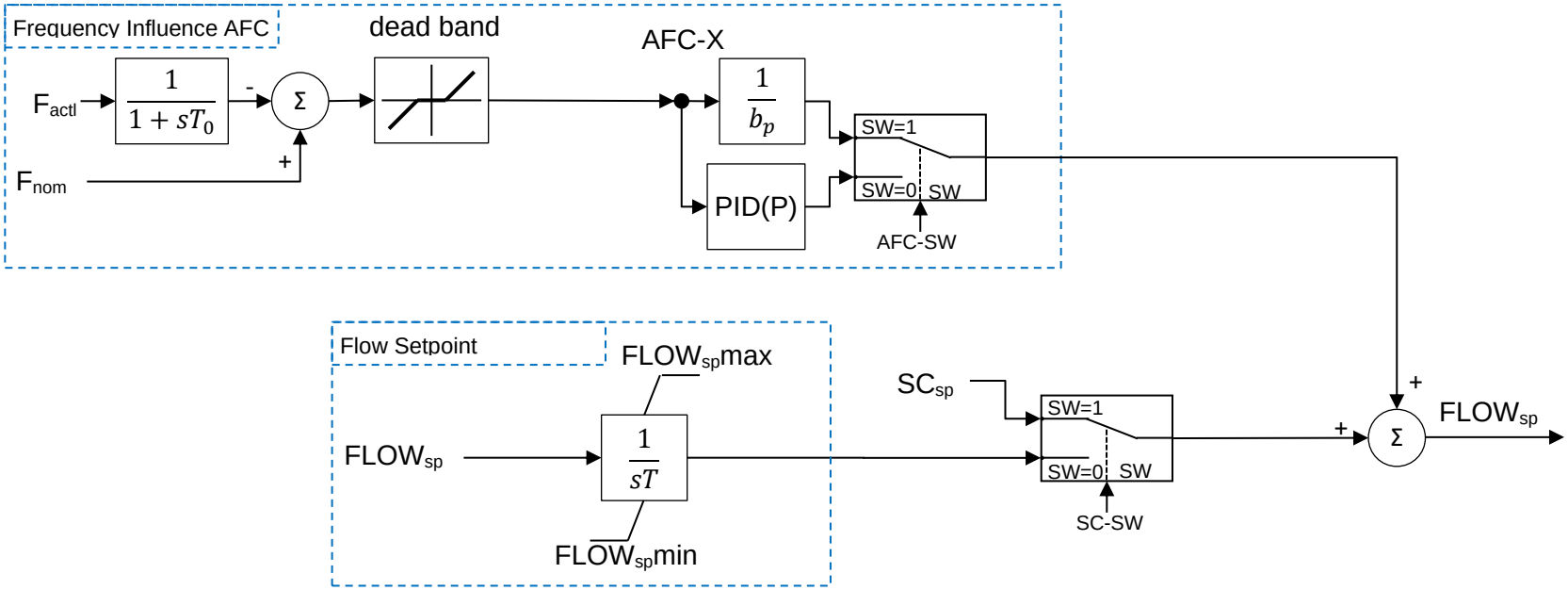
Transfer function for evaluation of the response to frequency disturbances for frequency influence with static droop

$$\frac{POW\text{-}FLOW_{sp}}{AFC\text{-}X} = \frac{1}{AFC\text{-}b_p} * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_i}{s} + \frac{s * POW\text{-}K_d}{1 + s * POW\text{-}T_d}\right)$$

Transfer function for evaluation of the response to frequency disturbances for frequency influence with PID

$$\frac{POW\text{-}FLOW_{sp}}{AFC\text{-}X} = \frac{AFC\text{-}K_p + \frac{AFC\text{-}K_i}{s} + \frac{s * AFC\text{-}K_d}{1 + s * AFC\text{-}T_d}}{1 + AFC\text{-}b_p * \left(AFC\text{-}K_p + \frac{AFC\text{-}K_i}{s}\right)} * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_i}{s} + \frac{s * POW\text{-}K_d}{1 + s * POW\text{-}T_d}\right)$$

## 7.3 Flow Control Mode



| Parameters | | Parameter Range | Typical | Comments |
|---|---|---|---|---|
| $FLOW_{sp}min$ | Minimum allowed Flow set point | 0 – 100% | 0% | |
| $FLOW_{sp}max$ | Maximum allowed Flow set point | 0 – 100% | 100% | |
| *FLOW-T* | Setpoint ramp time 0%-100% for Flow Control mode [s] | 10 – 120s | 60s | |
| $AFC\text{-}T_0$ | Speed Measurement Filter Time Constant [s] | 0.02 – 1s | 0.02s | |
| $AFC\text{-}b_p$ | Frequency influence (droop) bp | 0.001 – 0,1<br>(0.1% – 10%) | 0.05 (5%) | |
| AFC-SW | Frequency influence software switch | 0 – 1 | 1 | 0 = PID<br>1 = bp |
| SC-SW | Synchronous condenser software switch | 0 – 1 | 0 | 0 = Off<br>1 = On |
| $SC_{sp}$ | Flow set point in Synchronous condenser mode | 0 – 100% | 0% | |
| Flow control mode parameters | | | | |

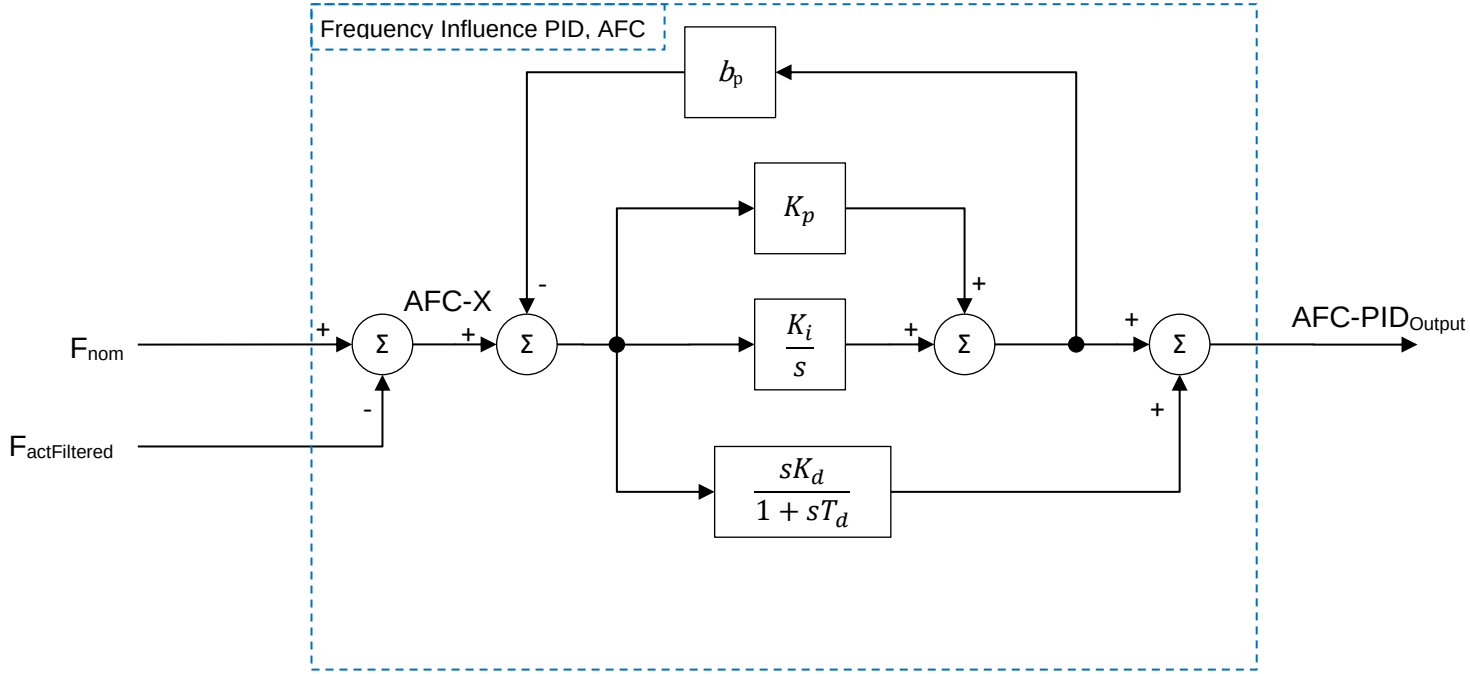
Transfer function for evaluation of the response to frequency disturbances for frequency influence with static droop

$$\frac{\mathrm{FLOW_{sp}}}{AFC\text{-}X} = \frac{1}{AFC\text{-}b_p}$$

Transfer function for evaluation of the response to frequency disturbances for frequency influence with PID

$$\frac{\mathrm{FLOW_{sp}}}{AFC\text{-}X} = \frac{AFC\text{-}K_p + \frac{AFC\text{-}K_i}{s} + \frac{s * AFC\text{-}K_d}{1 + s * AFC\text{-}T_d}}{1 + AFC\text{-}b_p \left(AFC\text{-}K_p + \frac{AFC\text{-}K_i}{s}\right)}$$

## 7.4 Frequency Influence PID loop

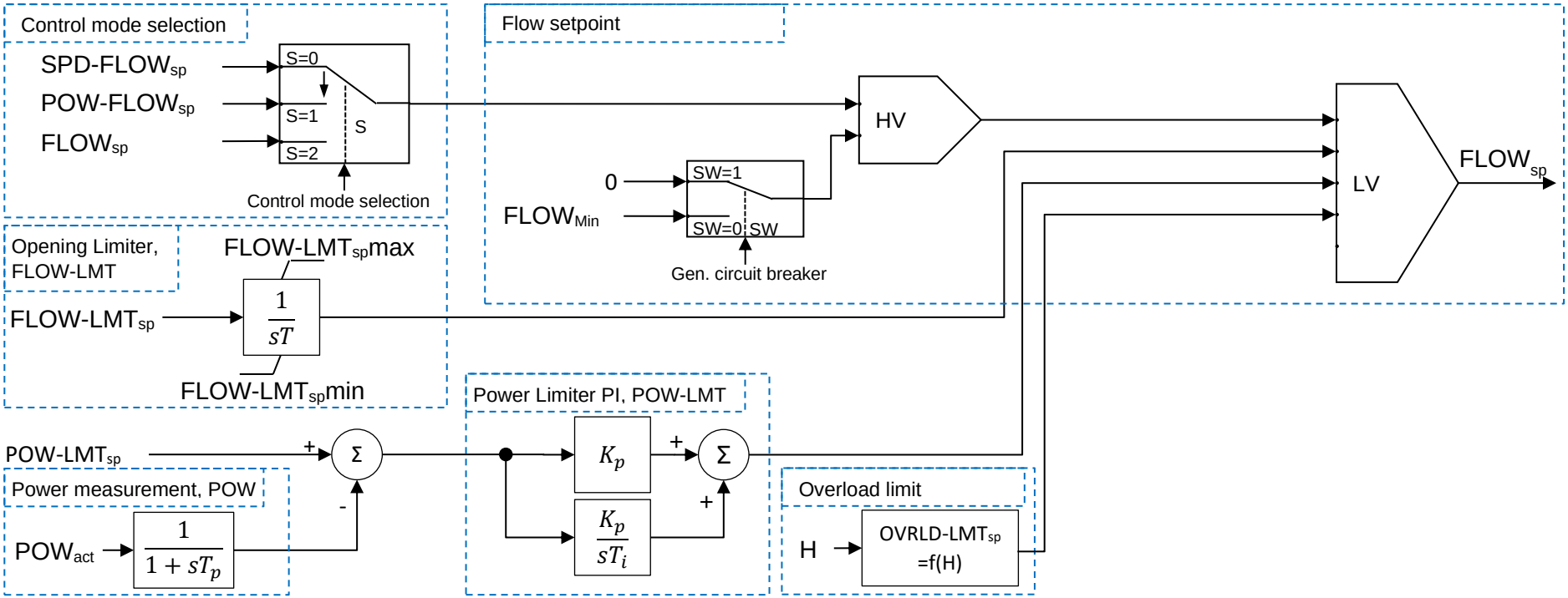


| Parameters | | Parameter Range | Typical | Comments |
|---|---|---|---|---|
| $AFC\text{-}bp$ | Permanent droop | 0 – 0.1 (0% - 10%) | 0.05 (5%) | |
| $AFC\text{-}K_p$ | Proportional gain | 0.1 – 20 | 1.5 | |
| $AFC\text{-}K_i$ | Integral gain | 0.01 – 100 | 0.5 | |
| $AFC\text{-}K_d$ | Derivative gain | 0 – 10 | 0 | |
| $AFC\text{-}T_d$ | Time constant of the damping device [s] | 0.1 – 10s | 0s | |
| Frequency influence PID parameters | | | | |

Transfer function Frequency Influence PID:

$$\frac{AFC\text{-}PID_{Output}}{AFC\text{-}X} = \frac{AFC\text{-}K_p + \frac{AFC\text{-}K_i}{s} + \frac{s * AFC\text{-}K_d}{1 + s * AFC\text{-}T_d}}{1 + AFC\text{-}b_p\left(AFC\text{-}K_p + \frac{AFC\text{-}K_i}{s}\right)}$$

## 7.5 Control signal selection



| Parameters | | Parameter Range | Typical | Comments |
|---|---|---|---|---|
| FLOW-LMT$_{sp}$min | Minimum allowed Flow Limiter set point | 0 - 100% | 0% | |
| FLOW-LMT$_{sp}$max | Maximum allowed Flow Limiter set point | 0 - 100% | 100% | |
| *FLOW-LMT-T* | Setpoint ramp time 0%-100% for Flow Limiter [s] | 10 – 120s | 60s | |
| *POW-T$_p$* | Power Measurement Filter Time Constant [s] | 0.02 – 1s | 0.2s | |
| FLOW$_{Min}$ | Minimum flow with generator circuit breaker closed | 0 – 10% | 2% + $O_{NL}$ | |
| *POW-LMT-K$_p$* | Proportional gain – Power Limiter | 0.1 – 20 | 0.8 | |
| *POW-LMT-K$_i$* | Integral gain – Power Limiter | 0.01 – 100 | 0.15 | |
| POW-LMT$_{sp}$ | Maximum Power Limit | 0 - 100% | 100% | |
| OVRLD-LMT$_{sp}$ | Corresponding overload flow limit to Head H | - | | 11 interpolated points |
| H | Actual head | - | | |
| LV | Low value (Lowest value is transferred to output) | - | | |
| HV | High value (Highest value is transferred to output) | - | | |
| Control signal selection parameters | | | | |